

William Lu

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Education

University of Illinois Urbana-Champaign (UIUC) May 2027

B.S. in Mathematics & Computer Science GPA: 4.0/4.0

University of Wisconsin Madison May 2024

Transferred GPA: 4.0/4.0

Relevant Course: Machine Learning, Applied Parallel Programming, Algorithm, Computer Systems, Data Structures

Skills

Languages: Mandarin (Native), English (Fluent), Japanese (Functional)

Technical: C/C++, Python, Java, Unity, Git, PyTorch, Django, HTML, Javascript, CSS, LaTeX, CUDA

Projects & Experience

TikTok - Effect Core SDK Intern San Jose, CA
AI Agent Engineer May 2026 – Present

- Built a LangGraph-based multi-agent system to automatically analyze performance trace files using Trace Processor, identify performance bottlenecks and root causes, and localize issues within local code repositories.
- Developed an optimization planning and execution agent that translated profiling reports into actionable optimization strategies, performed code-level modifications, and validated improvements through profiling feedback.

Convolutional Layer CUDA Kernel Optimization Champaign, IL
CUDA Developer January 2026 – May 2026

- Implemented and optimized Convolutional Layer as matrix multiplication CUDA kernel, and using Tensor Core, shared memory tiling, constant memory caching, loop unrolling, CUDA streams, etc.
- Utilized Nsight Compute and Nsight Systems to analyze occupancy, memory access patterns, and Tensor Core utilization, iteratively optimizing GPU performance.

Job Application Management Web App Champaign, IL
Backend Engineer September 2025 – December 2025

- Collaborated in a full-stack team environment with backend-focused development and local containerized workflows to develop a full-stack web application that helps users track job applications.
- Built RESTful APIs in Django with well-structured data models and serializers to support scalable CRUD operations across multiple resources.
- Modeled relational database schemas using SQLite, enforcing data integrity through foreign keys and constraints.

Self-Supervised Representation Learning with SimCLR (PyTorch) Personal
Independent Developer August 2025 – October 2025

- Implemented a SimCLR (Simple Framework for Contrastive Learning of Visual Representations) pipeline from scratch in PyTorch, including data augmentation, a ResNet encoder, projection head, NT-Xent loss, and evaluation.
- Explored scalability issues (GPU memory limits, large batch size) and applied techniques like gradient accumulation and efficient data loading.

Tiny-Chat Virtual
Programmer & Assistant August 2025 – September 2025

- Coded for a multi-agent conversational environment framework, which allows large language models to simulate social interactions such as negotiation, persuasion, and collaboration.
- Implemented data loading and random sampling of agent profiles from databases (Local/HuggingFace datasets), and be able to save conversation to JSON file.

Unix-like Kernel Virtual
Programmer & designer May 2025 – August 2025

- Built a Unix-like kernel from scratch using RISC-V and C in a 3-person team, implementing multitasking, file system, virtual memory, etc.
- Deployed a 2-way associative cache, enabled process management and cross-process communication via a pipe. Wrote drivers for different devices such as block device and entropy device.

Personal Website Development

Independent Developer

Personal
August 2024 – Present

- Designed and built a personal website from scratch using HTML, CSS, and JavaScript.
- Deployed the site via GitHub Pages to demonstrate academic background, projects, and resume in a responsive, user-friendly portfolio format.

Minerva Game Studio

Game Programmer

Madison, WI
September 2023 – March 2025

- Coded for a student-run studio to develop a stand-alone roguelike game using Unity Engine, whose demo is released on steam.
- Designed and built skills (ability of game characters) and assisted implementing language localization to enhance gameplay and improve player adaptability.

Cathaypath Institute of Science Research

Research Group Leader

Virtual
April 2021 – August 2021

- Compared various Deep Learning models' ability in detecting one's mental health by analyzing social media posts.
- Led a student research group of 4 and co-published a paper on IEEE as a primary writer on Deep Learning's capacity in supporting people's mental health care.

Leadership

Association for Computing Machinery: SIGPwny

Active Member

Champaign, IL
September 2024 – March 2025

- Practicing Cybersecurity and Hacking skills in a Special Interest Group that focuses on information security under the Association for Computing Machinery at UIUC.
- Cooperate with teammates in competing the Catch The Flag competitions using cybersecurity knowledge such as git, pwn, cryptography, etc.

Eduphoria EdTech Co. Ltd.

General Assistant

Madison, WI
September 2023 – May 2024

- Organized events such as info sessions and ted talks in Madison campus for newly founded Chinese Company focusing on international education.
- Edited videos using CapCut for the Marketing Department, and collected data for the IT department in order to expand influence and provide better service for students.

National High School Game Academy

Game Programmer

Pittsburgh, PA
June 2022 – August 2022

- Applied various disciplines of game design such as narrative, sound, art, and coding to make video games, which visualized the possibilities of Human Computer Interaction with Virtual Reality (VR) in the video game industry.
- Created 2 video games in teams: a remake of the classic arcade game Centipede and VR Archery game using Unity Engine and HTC VIVE pro 2 (headset).

Game Design Club

Founder & Leader

Wheaton, IL
September 2021 – June 2023

- Established a student club that focused on creating stand-alone computer video games, and hosted weekly meeting in which taught Python, C#, and Unity to members.
- Created a 2D Maze Game, 2D Platformer Game (like Mario), and developed a 3D game demo that utilized OpenAI API in non-player characters (NPC) to explore AI's capabilities to make NPCs more realistic and make the game more immersive.